

Chapter 1

Background

This chapter develops the theory required to follow the rest of the thesis. We proceed from the general framework of sequential decision making under uncertainty (§1.1–§1.2), through the specific learning algorithms used to train our agents (§1.3) and the reward machinery that shapes them (§1.4), to the model-based world models that motivate the belief-centric view (§1.5). We then introduce the interpretability tools that this work rests on: the representational hypotheses that make internal states legible (§1.6), the family of activation-steering interventions we study (§1.7), and finally the belief/behaviour distinction that frames our central question (§1.8).

1.1 Reinforcement Learning and Markov Decision Processes

Reinforcement learning (RL) formalises an agent that learns to act by trial and error so as to maximise a scalar reward (19). The standard model is the *Markov Decision Process* (MDP), a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel $P(s' \mid s, a)$, reward function $R(s, a)$, and discount factor $\gamma \in [0, 1)$. At each step t the agent observes state s_t , samples an action $a_t \sim \pi(\cdot \mid s_t)$ from its *policy* π , receives reward $r_t = R(s_t, a_t)$, and transitions to $s_{t+1} \sim P(\cdot \mid s_t, a_t)$. The objective is to maximise the expected discounted return

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k}, \quad J(\pi) = \mathbb{E}_{\pi}[G_0]. \quad (1.1)$$

The discount γ trades off immediate against future reward and keeps the sum finite. Two value functions summarise long-run desirability: the *state value* $V^{\pi}(s) = \mathbb{E}_{\pi}[G_t \mid s_t = s]$ and the *action value* $Q^{\pi}(s, a) = \mathbb{E}_{\pi}[G_t \mid s_t = s, a_t = a]$. They satisfy the Bellman consistency equations, e.g.

$$V^\pi(s) = \mathbb{E}_{a \sim \pi, s' \sim P} [R(s, a) + \gamma V^\pi(s')]. \quad (1.2)$$

The *advantage* $A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$ measures how much better action a is than the policy’s average behaviour in s . For a parametric policy π_θ , the policy-gradient theorem gives an unbiased ascent direction on J :

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{\pi_\theta} [\nabla_\theta \log \pi_\theta(a_t | s_t) A^{\pi_\theta}(s_t, a_t)]. \quad (1.3)$$

Equation (1.3) is the basis of the actor–critic methods used in this thesis: a *critic* estimates the advantage and an *actor* π_θ is nudged to increase the log-probability of advantageous actions.

1.2 Partial Observability and Belief States

In most realistic environments the agent does not see the full state. This is captured by the *Partially Observable MDP* (POMDP) $(\mathcal{S}, \mathcal{A}, P, R, \Omega, O, \gamma)$, which augments the MDP with an observation space Ω and an observation model $O(o | s)$ (10). The agent receives only $o_t \sim O(\cdot | s_t)$, which need not identify s_t . A policy that conditions on the current observation alone is therefore insufficient; optimal behaviour requires integrating the *history* $\tau_t = (o_1, a_1, \dots, a_{t-1}, o_t)$.

The central object is the *belief state*, the posterior over hidden states given the history,

$$b_t(s) = \Pr(s_t = s | \tau_t), \quad (1.4)$$

updated recursively by Bayesian filtering,

$$b_t(s') \propto O(o_t | s') \sum_{s \in \mathcal{S}} P(s' | s, a_{t-1}) b_{t-1}(s). \quad (1.5)$$

The belief b_t is a *sufficient statistic* of the history for predicting the future: it is the minimal quantity from which the reward-relevant distribution over states can be recovered (3). Crucially, a POMDP over $\mathcal{S}$ is equivalent to a fully observable *belief MDP* over the continuous belief simplex $\Delta(\mathcal{S})$, so an optimal Markov policy $\pi^*(a | b)$ exists as a function of the belief alone. This equivalence is the conceptual backbone of the thesis: *the belief is the internal variable that must be inferred, maintained, and acted upon*, and everything we later probe or steer is an approximation to it.

Exact belief tracking is intractable for large $\mathcal{S}$. In deep RL the belief is approximated by a learned recurrent summary of the history. A recurrent network with hidden state $h_t \in \mathbb{R}^d$ implements

$$h_t = f_\theta(h_{t-1}, o_t, a_{t-1}), \quad \pi_\theta(a_t | \tau_t) = \pi_\theta(a_t | h_t), \quad (1.6)$$

with h_t trained end-to-end to carry exactly the history information the policy and value heads need. We use the Gated Recurrent Unit (GRU) for f_θ (5), whose gating stabilises credit assignment over long horizons. Under this view

h_t is a *learned, distributed belief*: it is not guaranteed to equal the Bayesian posterior (1.4), but a well-trained agent must encode within h_t whatever latent structure its optimal policy depends on. Recovering that structure from h_t is the subject of §1.6.

1.3 Proximal Policy Optimization

We train recurrent policies with Proximal Policy Optimization (PPO) (18), a first-order policy-gradient method that is stable and sample-efficient enough for POMDP navigation. PPO improves on the raw gradient (1.3) in two ways.

First, advantages are estimated by *Generalized Advantage Estimation* (GAE), an exponentially-weighted average of n -step temporal-difference errors that trades bias for variance via $\lambda \in [0, 1]$ (17):

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}, \quad \delta_t = r_t + \gamma V_{\phi}(s_{t+1}) - V_{\phi}(s_t), \quad (1.7)$$

where V_{ϕ} is the learned critic. Second, PPO constrains each update so the new policy stays close to the data-collecting policy $\pi_{\theta_{\text{old}}}$, which prevents destructively large steps. Writing the importance ratio $\rho_t(\theta) = \pi_{\theta}(a_t | s_t) / \pi_{\theta_{\text{old}}}(a_t | s_t)$, PPO maximises the clipped surrogate

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]. \quad (1.8)$$

The clip range ϵ caps the incentive to move ρ_t far from 1. The full objective adds a value-regression term and an entropy bonus,

$$L(\theta, \phi) = L^{\text{CLIP}}(\theta) - c_v \mathbb{E}_t [(V_{\phi}(s_t) - \hat{G}_t)^2] + c_e \mathbb{E}_t [\mathcal{H}[\pi_{\theta}(\cdot | s_t)]], \quad (1.9)$$

where $\hat{G}_t = \hat{A}_t + V_{\phi}(s_t)$ is the value target and $\mathcal{H}$ denotes entropy. The entropy coefficient c_e controls exploration; as we show in the experiments, it is decisive when the task admits a lazy, belief-free local optimum, because escaping that optimum requires sustained exploration of belief-conditional behaviour. For recurrent policies, equations (1.7)–(1.9) are optimised over sequences: the hidden state (1.6) is carried through each rollout and reset at episode boundaries, so gradients flow through time via backpropagation through the unrolled GRU.

1.4 Reward Shaping and Potential-Based Invariance

Sparse rewards make credit assignment hard: if reward arrives only at a distant goal, the gradient (1.3) is uninformative for most of an episode. *Reward shaping*

adds an auxiliary signal $F(s, a, s')$ to densify learning. The danger is that a poorly chosen F changes what is optimal, so the agent learns to exploit the shaping instead of solving the task. Ng et al. (13) show that this is avoided *if and only if* the shaping is *potential-based*: for some potential function $\Phi : \mathcal{S} \rightarrow \mathbb{R}$,

$$F(s, a, s') = \gamma \Phi(s') - \Phi(s). \quad (1.10)$$

Under (1.10) the set of optimal policies is provably unchanged, because the added term telescopes: over any trajectory the shaping contributes only $\gamma^T \Phi(s_T) - \Phi(s_0)$, a quantity that depends on the endpoints, not on the path taken. Potential-based reward shaping (PBRS) therefore accelerates learning without biasing the solution.

In this thesis the potential is derived from a *cost-to-go* oracle. Let $\text{ctg}(s)$ be the minimum number of primitive actions needed to reach the goal from s , computed by a shortest-path search on the (known) map; we set

$$\Phi(s) = -\text{ctg}(s). \quad (1.11)$$

Moving closer to the goal (ctg decreasing) yields positive shaping, providing a dense navigational gradient. Two properties of PBRS matter for our analysis. First, because the shaping is invariant, it *cannot* manufacture a preference the terminal reward does not already encode; it can only make an existing preference easier to discover. Second, its contribution to the *return difference* between two candidate endpoints is bounded by $|\Phi(s_A) - \Phi(s_B)|$, which for a small ctg gap can be negligible relative to a large terminal bonus. We return to both points when explaining why shaping toward a target does not, on its own, force a partially observable agent to condition its behaviour on the correct latent belief.

1.5 World Models and the Recurrent State-Space Model

An alternative to learning h_t purely from the RL loss is to learn an explicit generative *world model* that predicts observations and rewards, and to train the policy inside the model’s imagined rollouts. DreamerV3 (9, 8) is the state-of-the-art instance and underlies the model-based agents in this work. Its core is the *Recurrent State-Space Model* (RSSM), which factorises the latent state into a deterministic recurrent path h_t and a stochastic latent z_t :

$$h_t = f_\theta(h_{t-1}, z_{t-1}, a_{t-1}) \quad (\text{recurrent model}) \quad (1.12)$$

$$\hat{z}_t \sim p_\theta(\hat{z}_t \mid h_t) \quad (\text{prior / transition}) \quad (1.13)$$

$$z_t \sim q_\theta(z_t \mid h_t, x_t) \quad (\text{posterior / filter}) \quad (1.14)$$

where x_t is the encoded observation. The *prior* p_θ (1.13) predicts the next latent from dynamics alone (enabling imagination), while the *posterior* q_θ (1.14) corrects it using the actual observation — an amortised, learned version of the

Bayesian filter (1.5). The joint latent $s_t = (h_t, z_t)$ is the model’s belief state. Decoder heads reconstruct the observation $\hat{x}_t$ and predict reward from s_t , and the whole model is trained by maximising an evidence lower bound of the form

$$\mathcal{L}(\theta) = \underbrace{\mathbb{E}_q[\log p_\theta(x_t | s_t)]}_{\text{reconstruction}} - \beta \underbrace{\text{KL}[q_\theta(z_t | h_t, x_t) \| p_\theta(\hat{z}_t | h_t)]}_{\text{dynamics / representation}}. \quad (1.15)$$

The actor and critic are trained on trajectories *imagined* by rolling the prior forward without observations, then decoded back to interpretable predictions. Two features of this architecture are central to the thesis. **(i)** The belief (h_t, z_t) is a single shared object consumed by multiple heads — the actor, the reward predictor, and the observation decoder — so there is no architecturally separate “skill” subspace. **(ii)** Because the decoder maps the latent back to observation space, the model exposes an explicit *belief readout*: one can decode what the agent “believes” the world looks like directly from s_t , and watch how that decoded belief changes under intervention. This makes the world model an unusually transparent substrate for the interpretability questions we now formalise.

1.6 Representations, Probing, and the Linear Hypothesis

Neural agents encode task-relevant variables in *distributed* representations: a concept is not a single unit but a pattern across many units (16). The empirically influential *linear representation hypothesis* (LRH) sharpens this: many high-level concepts are encoded as *directions* in activation space, so that the presence or value of a concept is (approximately) an affine function of the hidden state (12, 14). Under the LRH, a hidden state $h \in \mathbb{R}^d$ decomposes as a sum of concept directions plus residual, and the value of concept i is read off by projecting onto a direction v_i .

The standard tool for testing this is a *linear probe*: a simple classifier trained to predict a labelled latent variable y from a frozen activation h (1). For a categorical belief label with classes c , the probe is a linear-softmax map

$$\hat{p}(y = c | h) = \text{softmax}_c(Wh + b), \quad (1.16)$$

and high held-out accuracy is taken as evidence that the information y is linearly *decodable* from h . A key methodological caution, which this thesis makes concrete, is that decodability is a statement about what a *downstream reader* could in principle extract — it does *not* imply that the agent’s own policy head uses that direction. In our agents we attach such a probe as an auxiliary head trained jointly with the policy (a “belief head”); it provides a free, continuously-monitored readout of the latent belief without feeding back into the actor, so that decodability and use can be measured separately. When

a concept is linearly represented, a canonical estimate of its direction is the *difference of class means* in activation space,

$$v = \mu_+ - \mu_-, \quad \mu_{\pm} = \frac{1}{|\mathcal{D}_{\pm}|} \sum_{h \in \mathcal{D}_{\pm}} h, \quad (1.17)$$

where $\mathcal{D}_+$ and $\mathcal{D}_-$ are activations collected under the two values of the concept (11). The vector v points from one class centroid to the other and is the workhorse direction for the interventions in the next section. The LRH is, however, only approximate: *superposition* lets networks pack more features than dimensions by overlapping them in non-orthogonal directions (7), and some features occupy multi-dimensional subspaces rather than single lines. This non-orthogonality is precisely what makes targeted intervention leaky, a theme we develop in §1.8.

1.7 Activation Steering

Activation steering intervenes on a model’s computation at inference time by editing its hidden states along a chosen direction, rather than retraining weights. Given a steering direction v (often estimated by (1.17)), a layer activation h , and a scale α , the canonical edit is additive:

$$h' = h + \alpha v. \quad (1.18)$$

This is *activation addition* (ActAdd), which injects a contrast vector at inference to push behaviour toward or away from a target concept (20). *Contrastive Activation Addition* (CAA) constructs v as the mean difference of activations over matched positive and negative prompt pairs (15), i.e. a structured instance of (1.17). A complementary operation is directional *ablation*, which removes a concept by projecting it out of the activation,

$$h' = h - (\hat{v}^\top h) \hat{v}, \quad \hat{v} = v / \|v\|, \quad (1.19)$$

as used to suppress the refusal direction in language models (2). Finer-grained variants steer individual features discovered by *sparse autoencoders*, which decompose h into a sparse, ideally monosemantic dictionary and amplify or suppress single entries (4, 6). In this thesis we apply (1.18)–(1.19) to the recurrent belief carrier h_t of an RL agent: the direction v is a difference-of-means over the latent belief (e.g. activations on maps of one category minus another), and it is injected over a window of timesteps while the agent acts. The behavioural consequence — whether the intervention changes the agent’s downstream choice — is then read out from its actions, and the belief consequence from the decoded world model or belief probe.

Every method in this family shares an implicit assumption: that editing along v leaves the computations served by other directions untouched. Because representations are entangled and non-orthogonal (§1.6), any edit with a nonzero projection onto other functional directions perturbs them too. Whether a given

belief direction is in fact *used* by the policy, and whether steering it produces a clean behavioural change or leaks into unrelated capabilities, are empirical questions — and the ones this thesis is built to answer.

1.8 Belief, Skill, and the Decoupling Question

We can now state the conceptual frame precisely. We distinguish two readouts of a shared latent state z (either the GRU state h_t or the RSSM latent (h_t, z_t)). The *belief readout* $B(z)$ is the model’s inferred distribution over hidden world structure — e.g. the decoded observation of a world model, or the belief probe (1.16). The *policy readout* $\pi(z)$ is the distribution over actions. A latent variable is *behaviourally load-bearing* when the policy’s optimal action depends on it; it is *merely decodable* when $B(z)$ recovers it but $\pi(z)$ does not use it.

Localised steering implicitly assumes these are separable: that one can edit the belief without disturbing the policy, or edit the policy’s target behaviour without disturbing the belief. Following the diagnostic used in this work, an intervention δ applied to z is *belief-preserving* iff it moves the policy while leaving the belief readout intact:

$$\|B(z + \delta) - B(z)\| < \varepsilon \quad \text{and} \quad \|\pi(z + \delta) - \pi(z)\| > \eta, \quad (1.20)$$

for pre-specified tolerances ε (permitted belief drift) and η (required behavioural change). Symmetrically, a behaviour-targeting edit is *clean* only if it does not silently shift $B(z)$. When belief and policy readouts load onto overlapping directions of a shared latent, no strictly linear edit can satisfy (1.20) exactly; the severity of the resulting cross-talk is a property of the representational geometry, not of the method. This is why architectures with an explicit belief decoder are valuable: they let $B(z)$ be measured directly, turning (1.20) from a philosophical distinction into a checkable quantity.

The environments and agents in the following chapters are designed to make this question concrete and controllable. The agent must infer a latent map property (the belief) from a partial history and select among behaviours accordingly (the skill), while an auxiliary belief head (1.16) and, for the world-model agents, an observation decoder provide independent readouts of $B(z)$. This setup lets us ask, and answer empirically, the two questions that motivate the thesis: *(i)* when is a decodable belief actually used by the policy, and *(ii)* does steering the belief direction change behaviour as intended, or does it drift the belief while leaving behaviour untouched — or vice versa.

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